



Inspiring Excellence

BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: Smart Medicine Reminder Box with Unauthorized Access Safety
by

[Group_No-08]

[Section-04]

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Abstract

The project is aimed to address 2 key problems of home healthcare. Drug possession among patients and children having access to drugs illegally. It includes planning of embedded system based on Arduino microcontroller, RTC to keep time properly, and a locking mechanism of the servo controller. It is a system that is able to unlock at the required times, it also gives notifications about the medicines, an alarm is released in case of unwarranted access. It emphasizes the development of viable meaningful change in patient safety and tranquility.

Keywords

Embedded system, Arduino, Home Healthcare, Automation.

1. Introduction

Problem and motivation: Sometimes Elderly people can forget about taking the doses or take the wrong doses of medicine. In the homes, loose medication is incredibly dangerous to children and thus it causes unintentional poisoning. Standards pillbox does not contain active reminders or security.

Objective(s) of the project:

1. System consisting of a machine that will lock, unlock a medicine compartment according to a time based on a programmable time.
2. Have a security system that indicates tempering.
3. Making sure that harmful medicine does not fall in the wrong hands as well as children.
4. Invent an appliance without an internet.

The design and development of an embedded system that links hardware sensors and software control are part of the scope of this project. It is done by connecting a microcontroller with various peripherals to monitor real-time events like

medication timetable and improper access to the box. The project puts more focus on the creation of an independent, offline system that will provide sound and visual feedback to the users.

The project represents the major ideas behind computer interfacing by making use of the I2C communication to interface the Arduino with the RTC and the LCD with the fewest pins. It also shows the interface of GPIO by accepting and sending digital signals to buttons and switches and sending signals to LEDs and a buzzer. Also, the interfacing with the RTC demonstrates the time based data processing. Besides, a simple human machine interface is built by the LCD and the buttons.

2. Related Work/ Inspiration

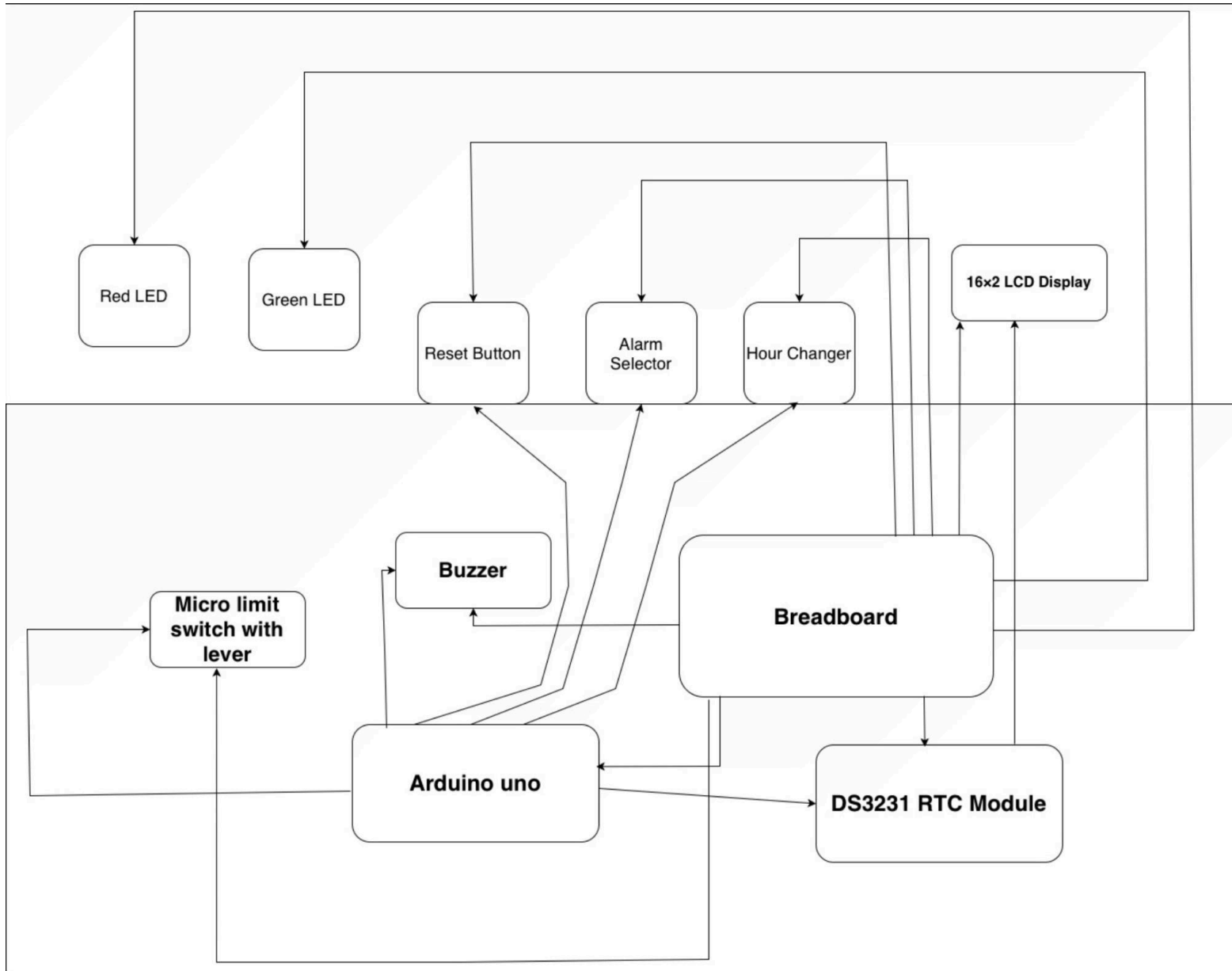
The current medication management systems in the market are either manual organizers or automated systems with high prices. Pill organizers that are done manually are inexpensive, but provide no form of notification and can not be secured against unauthorized access. On the one hand, expensive Wi-Fi networks and smartphone applications are needed in high-end automated pill systems. This may not be affordable to many in low bandwidth network environment. The justification of the project is to bridge a gap that exists between these two systems.

Our Smart Medicine Reminder Box is an enhanced version of the traditional technology, as it deals with active security notifications and independent operation. The system will operate fully off-line and can still be manipulated even when the internet goes offline. Instead of being a mere storage box, the system is active and tracks the times when it is used. It is able to distinguish between prescribed medication hour and unanticipated opening. In the event that an unauthorized person, like a child, opens the box when it is not on the schedule a loud warning buzzer is flown and immediately a red LED light is lit to attract attention of people

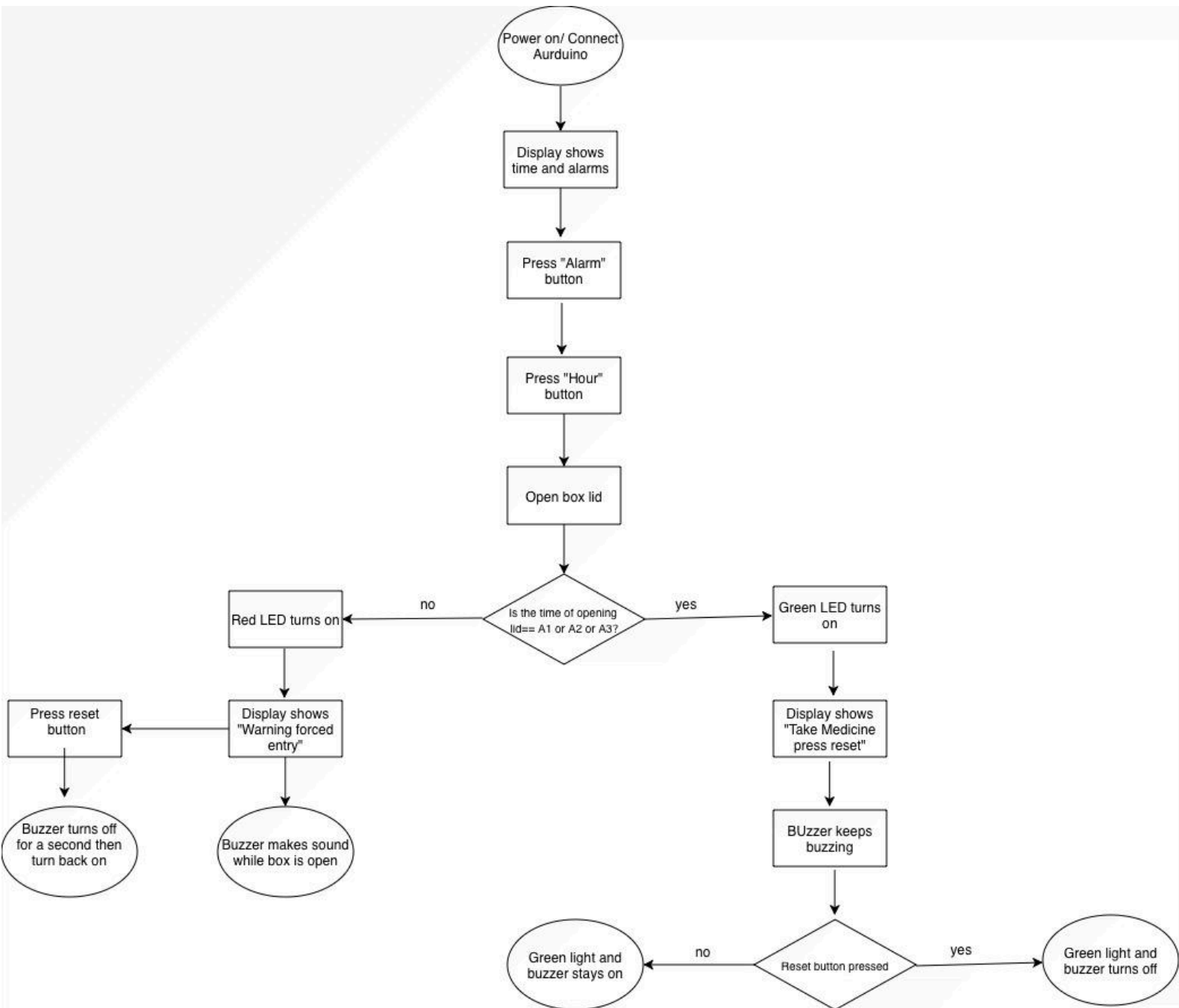
in charge. The fact that it has an inbuilt data logging feature is one of the most important benefits of this system. Such information gives a clear and trustworthy record that could be checked by the family members or doctors to ensure that drugs were used as per the schedule that was provided.

3. Technical Approach

System Architecture:



Fig# Block Diagram



Fig# Process Flowchart

Components Used:

1. **Arduino Uno R3:** The brains of the whole operation; it contains the code and corresponds to all other parts (such as the lock and alarm) on what and when to do what.
2. **DS3231 RTC Module:** This is the timekeeper of the system. It also has its small battery, and thus it is never confused of the actual time even when the main power is off.
3. **16x2 LCD Display (with I2C):** the front of the device. It notifies the user of the time, battery life, and reminds them of taking medicine, etc.
5. **Active Buzzer:** Voice of system. It rings to alert the patient that he should take pills or a loud siren to alert someone that he or she is attempting to break in.
6. **Momentary Push Buttons:** The buttons that allow you to manipulate the box, e.g. the menu to adjust an alarm time or to enter the emergency unlock code.
7. **LEDs (Green, Red):** Visual indicators: Light indicators that are similar to traffic lights: Green indicates that there is nothing wrong with the car, Red indicates that the car has been tampered with.
8. **Resistors (220 ohm):** Little shields to the LEDs. They restrict the supply of electricity to avoid cases of the lights exhausting their power.

10. Solderless Breadboard: This is the base in which we construct the circuit. It allows us to insert the wires and components with little difficulty to check the connection without having to use any glue and soldering.

11. Jumper Wires (Male to Male and Male to Female): The nervous system of the project. These electric wires are used to transfer the data and electric signals, and these wires connect the Arduino to the sensors and motors.

13. Micro limit switch containing lever: Identifies the state of the box lid whether it is open or closed.

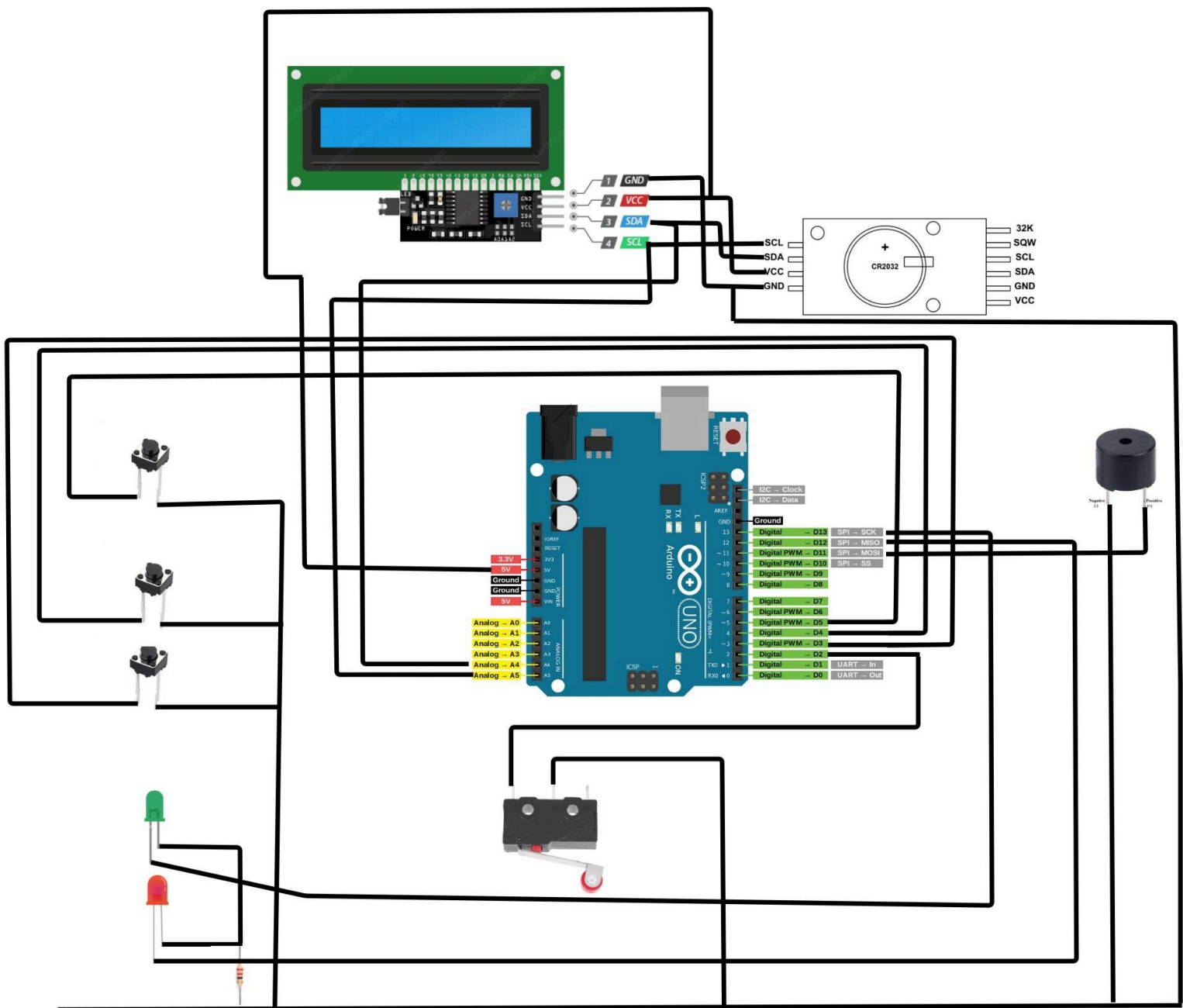
Cost Breakdown:

Product Name	Quantity	Unit Price (BDT)	Total (BDT)
Arduino Uno R3 (SMD) previous owned	1	550	550
DS3231 RTC Module (with Battery)	1	350	350
16×2 LCD Display with I2C Adapter	1	360	360
Active Buzzer - 5V	1	20	20
Momentary Push Buttons (Tactile)	4	6	24
LED (Red, Green - 5mm)	2	5	10
Resistor Pack (220 Ω)	10	2	20
Solderless Breadboard	1	140	140
Jumper Wires (M-M / M-F Bundle)	1	120	120
Micro Limit Switch with Lever	1	35	35
TOTAL			1,629

Communication Protocol: The protocol of communication in this system is I2C. It also enables the integration between the Master, which is the Arduino Uno and the two slaves, 16x2 LCD Display and the DS3231 RTC Module. We decided to use I2C since it uses only two signal lines SDA (Serial Data) and SCL (Serial Clock) and other components of the Arduino receive more free pins. This makes the circuit very easy and it can be used to connect other components such as the buzzer and the buttons.

The system employs ordinary general purpose input/output (GPIO) interfaces to real-time components since it is not a complicated serial protocol. Input Interfacing: Push Buttons and the Micro limit switch transmit digital information to the Arduino to turn off the buzzer and the unwarranted opening. Output Interfacing: In this case, Arduino makes use of digital messages to manipulate the running buzzer and the LEDs (Red and Green). These components work on high and low logic to give feedback to the user.

Circuit/Schematic:



Ground

Implementation: The application of the Smart Medicine Reminder Box commences when the Arduino Uno is connected to a DC power. When first booted the 16x2 LCD display will automatically boot up to display the current real-time and active alarm schedules given by the DS3231 RTC module. This system uses I2C protocol to provide communication among the microcontroller and these modules with a minimum amount of wiring. A micro limit switch is the initial sensor of knowing whether the box lid has been opened or closed. The opening of the lid in a programmed time when medication should occur causes the Arduino to activate the Green LED and the active buzzer to alert the patient. Instead, when the box is opened outside of these set periods, the system would regard the access as an unauthorized user; it will ring an alarm and turn on a Red LED to notify the caregivers about a possible safety hazard. All the status updates such as medication reminders and forced entry warnings are displayed on the LCD screen. Last but not least; the user may press the momentary reset button to discontinue the alarm and back the device to its monitoring mode.

Challenges:

1. Couldn't decide on power source; later used arduino to dc power source
2. Couldn't figure out how to use servo without external adapter, so we decided to exclude the lock system.

3. Placed the led's inside the box, but it wasn't visible so we soldered them in a way that they were visible from the outside of the box.
- 4.Placed the switches inside the box, but it wasn't visible so we soldered them in a way that they were visible from the outside of the box.
5. Current going through the leds were too high so we used 220 ohm for resistance to limit the current flow.

4. Sustainability & Impact

Sustainability:System incorporates a microcontroller logic with low power requirements. The servo motor is switched on only when the locked or the unlocked occurs. LCD backlight is used to provide an efficient system to use on battery saver. It uses parts that can be readily detached and reused in various projects thus minimizing e-waste. The system will help to minimise plastic waste in the long run by enabling its users to refill a robust container.

Impact: This enhances the autonomy of the elderly patients who are always watching the time of taking the medication. Active lock systems decrease directly the chances of children to touch the drugs by accident.

Further Development: Addition of ESP8266 to send miss dose notification directly to phone via WiFi. Switching the button system to fingerprint sensor to have more rapid and secure access. Use a stepper motor rotation system to have different pills at various periods of the day.

Limitations:

Hardware Limitations:

None is Locked physically: There is no physical locking system or a servo motor in the system.

Design shortcomings: A mechanical lock was not possible as the time and technical knowledge were lacking. We also operate this system using Arduino rather than a battery power source that is similar to lack of power.

Alert-Based Security The system exclusively relies on the buzzer and LEDs to indicate the occurrence of unauthorized access.

limitation on software or interfacing:

Fixed Alarm Time: There is no full customization in the program, and this is hardcoded to medication times.

Button Sync Problem: Buttons should not be used to change the preset alarm time, and it does not override these settings correctly.

Inadequate Storage The alarm data is not stored permanently and is restored to default values on restart.

5. Results & Discussion

Process	LED	Buzzer	Duration	Response time
Open the lid at the correct time	Green LED on Red LED off	Buzzing	Won't stop unless pressed override	0.5s
Open the lid at the incorrect time	Red LED on Green LED off	Buzzing	Won't stop unless pressed override	0s
Open the lid at the incorrect time (emergency)	Red LED on Green LED off	Buzzing	Won't stop unless pressed override	0s
Press the override/emergency button	Both LED off	Buzzing off	1s	0s

The experimental results prove that the system is very responsive and the response time is 0s to 0.5s. This performance is related to our project objectives directly in the following aspects:

1. **Time Accuracy Achievement:** DS3231 RTC module is utilized to trace the medication hours schedule. Once the box is opened at the right moment, a quick

feedback is given by the Green LED and Buzzer that will help the elderly patients to administer the correct dosage at the right time.

2. Security and Tamper Detection: We were expected to install a security system to identify tampering. The findings indicate that the Red LED and the Buzzer are ignited when any unauthorized opening is done immediately (0s). We wanted to physically lock the servo but because of time and lack of knowledge, we were not able to do it. This alarm-style security is able to fulfill the objective of telling the seniors about unauthorized access or danger to children's security.

3. Reliability and Practicality: The findings demonstrate that the system is 100% reliable in offline settings. With the efficient communication through I2C and interfacing with theGPIO, we developed a system that would offer high domestic safety at an affordable cost without the use of complex internet-based systems.

6. Conclusion

The Smart Medicine Reminder Box has managed to offer an affordable and reliable solution to home healthcare. The system will consider active monitoring and real-time notifications through which any unauthorized access will be revealed immediately by a security buzzer. This makes it very safe and does not require any sophisticated mechanism of locking up the medicine. The project will ensure that the elderly patients will take their medicines at the right time and that a record is kept. This record is easily checked by family members and doctors so as to ensure that the treatment is undertaken appropriately. The device does not need the internet to work therefore it is reliable and can be used in any household easily.

All in all, this project addresses the fact that embedded systems can be used to address real-world challenges through the integration of sensor-based security and user-friendly design. It is an economical and long-term substitute to costly commercial dispensers that enhance domestic safety and patient care by a long margin.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Avrow Jothi Bose	22299227	Components collect+Building	Bought all the components and built the primary setup of the project and flowcharts.
Pallab Saha	22299383	Report Writing	Did research about the communication protocol and wrote the report from the beginning to the end.
S.M Shihabur Rahman Ratul	22201177	Report Writing	Research and writing, workflow , scope, limitations, results and discussions.
Tanvir Bashar Aurpo	22201903	Circuit Building + Software interface	Research about how DS3231 RTC module and how it can

			be connected to arduino to store data (alarm info) and display the real time clock in I2C display Code troubleshooting and implementation
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References

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